

SOFIA, Bulgaria

WMO: 156140

Lat: 42.695N

Lon: 23.406E

Elev: 1742

StdP: 13.79

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.0	14.3	4.8	7.5	13.0	9.3	9.4	17.5	24.3	31.8	22.1	31.4	3.6	270	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.5	91.6	66.6	87.9	65.9	85.2	65.4	69.9	83.8	68.4	81.6	67.2	80.1	7.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.9	102.0	74.7	64.3	96.2	73.6	62.7	91.0	72.0	34.9	83.9	33.7	82.0	32.6	79.8	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.9	18.1	15.7	1.0	97.3	6.3	4.2	-3.6	100.3	-7.3	102.7	-10.9	105.1	-15.5	108.1		
(5)			0.4	73.2	6.3	2.4	-4.1	74.9	-7.7	76.3	-11.2	77.6	-15.8	79.3	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.0	30.5	35.4	43.1	52.0	60.4	67.0	71.1	71.2	63.0	53.1	43.1	33.0
	DBStd	15.88	8.62	9.09	7.80	6.96	5.86	5.85	5.10	5.14	6.34	7.15	7.91	8.47
	HDD50	2126	605	411	236	56	3	0	0	0	1	52	231	531
	HDD65	5324	1070	828	678	391	164	45	8	10	110	371	656	993
	CDD50	2857	0	3	23	117	325	510	655	657	391	148	25	3
	CDD65	581	0	0	0	2	21	105	198	202	50	3	0	0
	CDH74	6081	0	0	3	47	266	1037	2020	2079	573	55	1	0
	CDH80	2182	0	0	0	5	40	325	798	835	173	6	0	0
Wind	WSAvg	5.7	5.4	6.3	6.7	6.4	5.9	5.5	5.6	5.2	5.3	5.1	5.4	5.3
Precipitation	PrecAvg	24.7	1.3	1.4	1.6	2.0	2.9	3.0	2.4	2.0	1.9	1.7	1.7	1.9
	PrecMax	42.2	3.7	3.3	4.6	5.9	5.8	6.4	7.4	7.5	7.8	4.9	3.9	6.7
	PrecMin	13.1	0.1	0.0	0.1	0.1	0.5	0.3	0.3	0.0	0.1	0.0	0.1	0.0
	PrecStd	5.9	0.9	0.7	0.9	1.1	1.4	1.6	1.6	1.6	1.6	1.3	0.9	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.1	64.6	71.7	79.1	84.3	91.8	97.0	95.2	89.9	79.1	69.7	61.2
		MCWB	47.5	50.7	52.3	58.0	61.4	67.8	65.9	66.5	64.2	60.8	56.0	51.0
	2%	DB	50.1	57.5	66.5	73.7	79.8	87.4	91.7	91.5	84.6	74.9	64.5	53.5
		MCWB	44.1	47.2	50.7	55.9	61.3	66.9	66.6	66.7	62.5	59.3	54.2	46.8
	5%	DB	46.3	53.3	61.9	68.9	76.9	84.0	87.7	87.9	80.6	70.0	59.3	48.4
		MCWB	42.2	45.0	48.5	53.9	60.7	66.5	66.4	65.7	61.7	57.4	51.7	44.6
	10%	DB	42.1	48.5	57.4	65.8	73.6	80.2	84.3	84.5	76.6	66.1	55.4	44.7
		MCWB	39.4	43.2	46.6	52.6	59.5	65.0	65.9	65.2	60.8	56.1	50.0	42.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.4	52.0	54.2	60.7	66.0	72.0	71.4	71.7	66.8	63.5	57.5	52.8
		MCDB	55.2	62.4	66.9	74.4	78.1	84.8	85.9	85.6	80.7	74.9	66.0	59.2
	2%	WB	44.9	48.3	51.6	57.8	63.8	69.2	69.7	69.6	65.0	60.9	54.8	47.8
		MCDB	49.2	56.3	63.6	69.9	75.2	81.7	83.7	83.9	78.2	71.1	62.5	52.5
	5%	WB	42.0	45.8	49.4	55.6	62.1	67.6	68.3	68.0	63.6	58.6	52.8	45.0
		MCDB	45.3	52.1	59.9	66.3	73.1	79.6	81.4	81.9	76.0	67.2	58.9	48.6
	10%	WB	39.3	43.3	47.5	53.6	60.4	65.9	67.1	66.6	62.1	57.0	50.3	42.1
		MCDB	41.7	48.2	56.4	62.3	70.8	77.6	80.0	80.3	73.3	64.5	55.0	44.3
Mean Daily Temperature Range	5% DB	MDBR	14.7	17.4	20.3	21.2	21.3	22.3	23.5	24.9	22.9	21.5	17.3	13.6
		MCDBR	20.3	26.0	30.2	28.6	27.1	28.1	29.4	30.4	30.5	29.2	24.9	19.6
		MCWBR	14.7	16.5	16.3	14.1	12.3	11.8	10.6	10.7	12.1	15.3	15.5	14.3
	5% WB	MCDBR	19.0	23.2	26.3	24.4	24.4	25.0	25.0	26.2	25.5	25.1	21.5	17.5
		MCWBR	14.3	15.3	14.8	12.7	11.9	11.6	10.6	10.5	11.3	14.0	13.7	13.2
Clear-Sky Solar Irradiance	taub	0.315	0.343	0.389	0.448	0.438	0.433	0.435	0.446	0.419	0.377	0.340	0.316	
	taud	2.361	2.296	2.221	2.063	2.162	2.215	2.209	2.179	2.229	2.337	2.410	2.394	
	Ebn at Noon	259	269	268	260	266	267	265	258	256	253	247	247	
	Edh at Noon	26	33	40	50	47	44	44	44	39	31	25	23	
All-Sky Solar Radiation	RadAvg	509	767	1091	1417	1691	1914	1982	1792	1319	909	576	427	
	RadStd	76	99	118	169	141	166	152	158	107	114	61	61	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+1.44	N/A	N/A	N/A	+0.90	+0.90	-261	-418	+239
(47) Regional (40 neighbors)		+0.88	N/A	N/A	N/A	+0.68	N/A	N/A	-250	N/A

Nomenclature: See separate page